



Lab: Bonus 1

Prepared by:
Abdelaziz Mohamed

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Objectives

The objectives of this lab are:

1. To be familiar with Ngspice.
2. To be familiar with AC analysis and Transient (tran) analysis.
3. To understand RLC circuits behavior.
4. To explore resonance concepts and step responses.

Part 1: Parallel RLC Circuit

Circuit Description

A parallel RLC circuit consists of a resistor (R), an inductor (L), and a capacitor (C) connected in parallel. The circuit exhibits harmonic oscillator behavior where energy oscillates back and forth between the inductor's magnetic field and the capacitor's electric field. The resistor acts as a damping element that dissipates energy.

At a specific frequency known as the resonance frequency (ω_0), the susceptance of the inductor and the capacitor cancel each other out. At this point, the total input impedance of the circuit becomes purely resistive, reaching its maximum magnitude, which is exactly equal to the resistance R . The phase angle of the impedance at resonance is zero.

Deliverables

A) Input Impedance Expression:

Find the analytical expression for the input impedance $Z_{IN}(s)$ of the parallel RLC circuit as a function of frequency.

B) AC Analysis ($|Z_{IN}|$ vs. Frequency):

Run an AC analysis to plot the magnitude of the input impedance, $|Z_{IN}|$, versus frequency $f(1G \rightarrow 20G)$ 60 points per decade. Identify and clearly mark the ranges of frequencies where the impedance is predominantly inductive and where it is predominantly capacitive. use the following values $L = 1nH$, $C = 1pF$, $R = 1K$.

C) Resonance Frequency Verification:

Find the resonance frequency analytically and compare it with the peak frequency observed in the simulation plot.

D) Quality Factor Evaluation:

Find the Quality factor (Q) analytically using the circuit parameters (R, L, C) and compare your analytical result with the Q extracted from the simulation.

E) Bandwidth (BW):

Find the bandwidth of the circuit. Hint : Bandwidth of parallel RLC is defined as the range of frequency at which the magnitude of the the $|Z_{IN}|$ falls to $R/\sqrt{2}$. also we find the BW as the ratio between the resonance frequency and the quality factor.

F) Parametric Sweep on R:

Run a parametric sweep on the resistor value $R = (500 : 100 : 1k)$ and plot $|Z_{IN}|$ overlaid on the same graph. Comment on the effect of changing R on the Quality factor (Q) and the sharpness of the resonance peak.

G) Transient response at Resonance:

Apply a sine wave transient current source at the exact resonance frequency. Plot the current passing through the capacitor (I_C) and the inductor (I_L). Comment on their phase relationship and amplitudes. Also, find the steady-state value of the output voltage. Hint : Make the amplitude of the current to be 1 A, How a passive circuit amplify the current :) ?!

H) Step Response:

Apply a step response to the circuit while running a parametric sweep on the resistor R . Plot the transient response and comment on the different damping behaviors observed

Part 2: Series RLC Circuit

Circuit Description

A series RLC circuit consists of a resistor (R), an inductor (L), and a capacitor (C) connected in series. The circuit exhibits harmonic oscillator behavior where energy oscillates back and forth between the inductor's magnetic field and the capacitor's electric field. The resistor acts as a damping element that dissipates energy.

At a specific frequency known as the resonance frequency (ω_0), the reactance of the inductor and the capacitor cancel each other out. At this point, the total input impedance of the circuit becomes purely resistive, reaching its minimum magnitude, which is exactly equal to the resistance R . The phase angle of the impedance at resonance is zero.

Deliverables

A) Input Impedance Expression:

Find the analytical expression for the input impedance $Z_{IN}(s)$ of the series RLC circuit as a function of frequency.

B) AC Analysis ($|Z_{IN}|$ vs. Frequency):

Run an AC analysis to plot the magnitude of the input impedance, $|Z_{IN}|$, versus frequency $f(1G \rightarrow 20G)$ 60 points per decade . Identify and clearly mark the ranges of frequencies where the impedance is predominantly inductive and where it is predominantly capacitive. use the following values $L = 1nH, C = 1pF, R = 1K$.

C) Resonance Frequency Verification:

Find the resonance frequency analytically and compare it with the minimum frequency observed in the simulation plot.

D) Quality Factor Evaluation:

Find the Quality factor (Q) analytically using the circuit parameters (R, L, C) and compare your analytical result with the Q extracted from the simulation.

E) Bandwidth (BW):

Find the bandwidth of the circuit. Hint : Bandwidth of a series RLC is defined as the range of frequency at which the magnitude of the $|Z_{IN}|$ rises to $R\sqrt{2}$. also we find the BW as the ratio between the resonance frequency and the quality factor.

F) Parametric Sweep on R:

Run a parametric sweep on the resistor value $R = (500 : 100 : 1k)$ and plot $|Z_{IN}|$ overlaid on the same graph. Comment on the effect of changing R on the Quality factor (Q) and the sharpness of the resonance dip.

G) Transient response at Resonance:

Apply a sine wave transient voltage source at the exact resonance frequency. Plot the voltages across the capacitor (V_C) and the inductor (V_L). Comment on their phase relationship and amplitudes. Also, find the value of the voltage at the capacitor. Hint : Make the amplitude of the voltage to be 1 V, How a passive circuit amplify the voltage :) ?!

H) Step Response:

Apply a step response to the circuit while running a parametric sweep on the resistor R . Plot the transient response and comment on the different damping behaviors observed